Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

Consider a trained SVM on n data points in $\mathbb{R}^{10}$, out of which only m $(m < n)$ are support vectors. Assume that after training, the weight vector w is already computed and stored. What is the time complexity of making a prediction for a new point $x \in \mathbb{R}^{10}$?

A. $O(1)$
B. $O(n)$
C. $O(m)$
D. $O(\log m)$

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #2 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

In the linearly non-separable case, what effect does the C parameter have on the SVM mode?

A. it determines how many data points lie within the margin

B. it is a count of the number of data points which do not lie on their respective side of the hyperplane

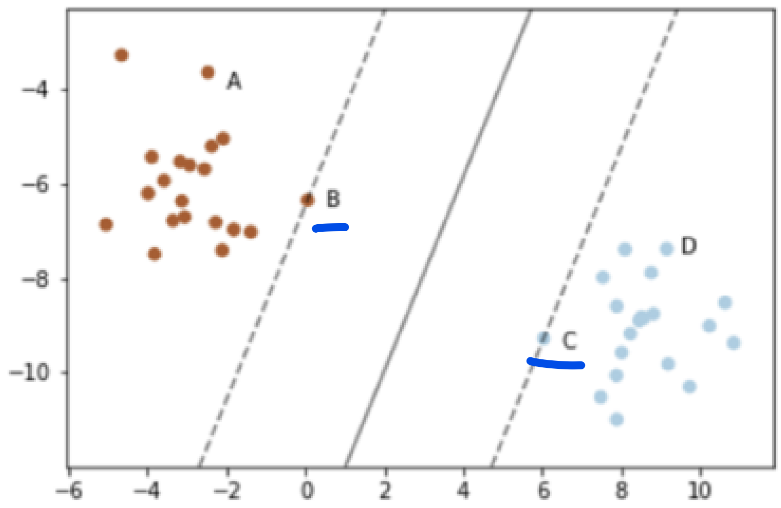
C. it allows us to trade-off the number of misclassified points in the training data and the size of the margin

D. it counts the support vectors

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #3 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

The margin of a support vector machine classifier is plotted below. Which of the identified points are the support vectors?



A. B

B. B and C

C. B, C, and D

D. A and C

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

A linear SVM has a weight vector $\vec{w} = [0, -c]$. If the margin width is k, determine the value of c.

A. $\frac{k}{2}$

B. $\frac{2}{k}$

C. $\frac{1}{k}$

D. $\frac{k}{\sqrt{2}}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 34sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset with positive points $A(-s, 0), C(t, 0)$ and a negative point $B(0, k)$. A hard-margin SVM classifier is used. What are the values of the weight vector $\vec{w}$ and bias b?

A. $\vec{w} = [0, -2/k], b = 1$

B. $\vec{w} = [0, -k/2], b = 1$

C. $\vec{w} = [-2/k, 0], b = k/2$

D. $\vec{w} = [0, -1], b = k/2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You train linear SVM and linear logistic regression on a dataset D, obtaining boundaries f_{SVM} and f_{LR}. Now form $D' = D \cup \{(x^*, y^*)\}$, where (x^*, y^*) is correctly classified by both models and lies far from the current boundary (large positive margin). Compared to the boundaries trained on D, what happens after adding (x^*, y^*) ?

A. The SVM boundary on D' is identical to its boundary on D, while the logistic regression boundary typically shifts slightly toward (x^*, y^*).

B. Both boundaries remain identical to those on D.

C. Both boundaries shift by a similar amount toward (x^*, y^*).

D. The logistic regression boundary is unchanged, while the SVM boundary shifts.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In a hard-margin SVM, the optimization problem is to minimize

$$J = \frac{1}{2}||w||^2$$

What are the additional constraints for this optimization problem?

A. $y_i\left(w^T x_i + w_0\right) \leq 1$ for all training points; this allows some misclassification.

B. $y_i\left(w^T x_i + w_0\right) \geq 1$ for all training points; this ensures all training points are correctly classified and outside the margin.

C. $y_i\left(w^T x_i + w_0\right) \geq 0$ for all training points; this only enforces correct classification but does not guarantee a margin.

D. No additional constraints are required; minimizing $||w||^2$ alone guarantees correct classification.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In an SVM, suppose the margin constraints are changed from

$$y_i\left(\mathbf{w}^T x_i + b\right) \geq 1$$

to

$$y_i\left(\mathbf{w}^T x_i + b\right) \geq \rho, \quad i = 1, 2, 3 \quad \text{for any } \rho \geq 1$$

Which of the following best describes the effect of this change?

A. The separating hyperplane itself changes, leading to a different decision boundary.

B. The separating hyperplane remains the same, but the parameters $\mathbf{w}, b$ scale with ρ.

C. Both the parameters and the decision boundary remain unchanged.

D. The SVM becomes invalid for $\rho > 1$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

After training a Support Vector Machine (SVM), each Lagrange multiplier λ_i takes either a zero or nonzero value. Which of the following statements is correct?

A. A non-zero λ_i indicates that example i is a support vector.

B. A zero λ_i indicates that example i is a support vector.

C. A non-zero λ_i indicates that the learning has not yet converged to a global minimum.

D. A zero λ_i indicates that the learning process has identified support for example i.

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Select Type Award: 1 Penalty: 0 Machine Learning

When solving a soft-margin SVM (C-SVM) for binary classification $\{-1, +1\}$, which of the following statements are true?

A. All support vectors are necessarily classified correctly.

B. Some support vectors may be classified incorrectly.

C. Support vectors are necessarily on the border of the maximum-margin hyperplane.

D. If $0 < \alpha_i < C$, then x_i is a support vector lying on the margin of the maximum-margin hyperplane.

Your Answer: Correct Answer: B;D Not Attempted Time taken: 00min 00sec Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

In the context of Support Vector Machines (SVMs), which of the following statements are correct regarding the effect of removing support vectors on the optimal margin?

A. Removing some constraints (support vectors) in a constrained maximization problem can only make the solution at least as good as before.

B. The set of support vectors is usually not unique; different training subsets can lead to the same separating hyperplane.

C. In general datasets, removing support vectors can cause the margin to decrease, depending on geometry.

D. In general, if you maximize a function over a smaller set, the value is never larger than maximizing over a bigger set. Symbolically:

$$\max_{x \in (A \cap B)} f(x) \leq \max_{x \in A} f(x)$$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following statements are true about Support Vector Machines (SVMs)?

A. Increasing the hyperparameter C tends to decrease the training error.

B. The hard-margin SVM is a special case of the soft-margin SVM with the hyperparameter C set to zero.

C. Increasing the hyperparameter C tends to decrease the margin.

D. Increasing the hyperparameter C tends to increase the number of support vectors.

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you have a dataset $(x_i, y_i) \in \mathbb{R}^d \times \{-1, 1\}$. After applying a feature transformation

$$\phi(x) = Ax + c, \quad A \in \mathbb{R}^{n \times d}, c \in \mathbb{R}^n$$

a hard-margin SVM achieves zero training error. What can we conclude about the original dataset?

A. The original dataset was not linearly separable, and the affine transformation made it separable.

B. The original dataset was already linearly separable.

C. Nothing definite can be concluded about the separability of the original dataset.

D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

x_1	x_2	y	α
1	0	1	1.11
2	1	1	0
0	1	-1	0.89
-1	-1	-1	0.22

Consider the data shown above, from which a linear hard margin SVM was obtained. The table includes the input features, x_1 and x_2, the class labels y, and the computed Lagrange multipliers of the solution, α.

Compute the value of the weight component w_1 of the obtained hard margin SVM.

A. -0.50

B. -0.66

C. 1.00

D. 1.33

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following training dataset for a linear Support Vector Machine (SVM) in $\mathbb{R}^2$:

x	y
$(0,1)$	$+1$
$(1,0)$	$+1$
$\left(\frac{1}{2},\frac{1}{2}\right)$	-1
$\left(\frac{6}{13},\frac{4}{13}\right)$	-1

Which of the following is the correct solution for the weight vector w of the maximum-margin SVM?

A. $w=(1,0)$

B. $w=(0,1)$

C. $w=(\frac{1}{2},\frac{1}{2})$

D. $w=(\frac{6}{13},\frac{4}{13})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss